

Trainings ▾

Install

Rear Gear Train

Note : Install three guide pins, Part Number 3164977, to improve alignment of the front cover and front seal to the gear housing and crankshaft.

Apply a 1.5 to 2.0 mm [0.06 to 0.08 in] wide bead of sealant, Part Number 3164070, to the block side of the front gear cover in the path illustrated.

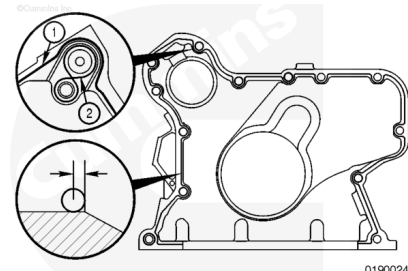
On engines equipped with an oil pressure sensor/switch located in the front gear cover, there are two critical sealant paths that **must** be followed:

1. In this area, the sealant **must** be applied towards the outer edge of the front gear cover to avoid an unused mounting hole in the cylinder block.

Note : To make sure the unused mounting hole in the cylinder block does not affect the sealing joint, fill the mounting hole with sealant, Part Number 3164070.

2. Sealant **must** be applied around both the mounting hole location and the oil supply hole in the front gear cover.

Note : Install the front cover within 10 minutes of applying the sealant, or the sealant will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before operating the engine.

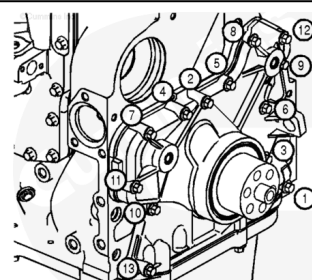


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Use the dowel rings to locate the front gear cover and install the front gear cover onto the cylinder block.

Install the front gear cover to the cylinder block with mounting capscrews. Tighten the front gear cover to the cylinder block mounting capscrews in the order indicated.

Torque Value: 24 n·m [212 in-lb]

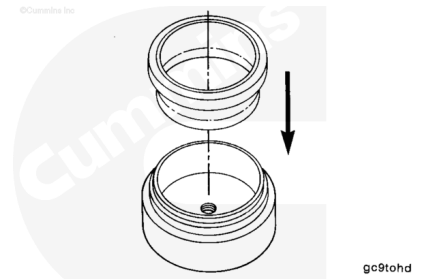


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Front Gear Train

Install a new front crankshaft seal into the front gear cover prior to installing the front gear cover. Refer to Procedure 001-023 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-023-tr.html)

Leave the plastic pilot installation tool in the front crankshaft seal.



⚠ CAUTION ⚠

The seal lip and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

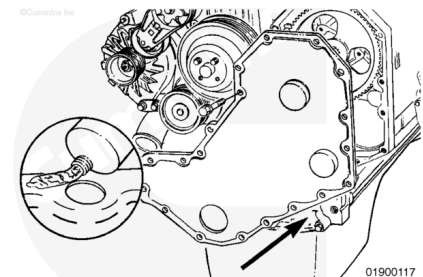
Stamped Steel Gear Cover - Apply a thin bead of sealant, Part Number 3164070, to the sealing surface of the front cover.

Note : Do not remove the plastic seal pilot tool from the front crankshaft seal at this time. Use the plastic seal pilot tool to guide the seal onto the crankshaft.

Note : Install the front cover within 10 minutes of applying the sealant, or the sealant will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before operating the engine.

Cast Aluminum Gear Cover - Place the gasket, Part Number 3918673, between the gear cover and the gear housing.

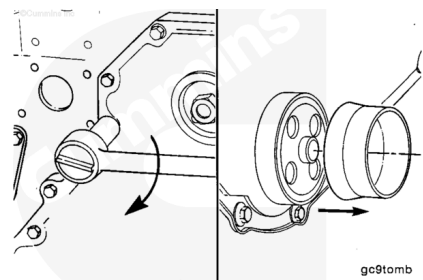
Install the front gear cover on the engine, with the plastic seal pilot tool to guide the front crankshaft seal onto the crankshaft.



Install and tighten the front gear cover capscrews.

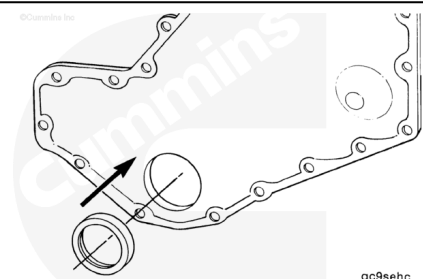
Torque Value: 24 n•m [212 in-lb]

Remove the plastic pilot tool from the crankshaft.



If previously equipped, or if the engine operates in a dusty environment, install a dust shield.

A pilot tool is **not** necessary for the dust shield. Slide the dust shield over the nose of the crankshaft.



Last Modified: 03-Aug-2016
